



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0474



Agricultural Machinery – Cassava Granulator CJ G 1103

Test Report Number	2018 – 0474
Date of Issuance	August 8, 2019
Valid Until	August 8, 2024
Agricultural Machinery	Cassava Granulator
Type	Engine-driven, Bladed drum-type, With chipper, Mobile, Hitch-type
Brand	CJ
Model	G 1103
Test Requested by	CJ DELFIN AGRI-INDUSTRIAL SALES National Highway Villarica, Midsayap. Cotabato

SUMMARY

Performance Criteria	Manufacturer's Specifications	AMTEC Test
Granulating Recovery, %, minimum	ND	NM
Granulating Capacity, kg/h	ND	2586.2
Noise Level, dB(A), maximum	90*	96.4

Note: *Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Department of Labor and Employment, Philippines. 2013.

OBSERVATIONS

1. Prime mover	6.3 kW KAWAMA Diesel Engine
2. Operator's manual	Not Provided
3. Previous valid test reports of similar brand and model	None
4. Manufacturer	CJ DELFIN AGRI-INDUSTRIAL SALES
5. Transmission system	V-belt and pulley
6. Loading of test materials	The hopper is accessible to the operator.
7. Transporting the machine	Manually transported
8. Cleaning of parts	Accessible
9. Adjusting and repair of parts	Not observed
10. Collecting output	Granulated cassava fell on the ground and was collected after testing.
11. Number of operators	3
12. Other remarks	No breakdown occurred. Provided with hitch point

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	April 20, 2018
Location of Test	National Highway Villarica, Midsayap. Cotabato
Items Tested	Input capacity, Noise level, Fuel consumption, and Number of operators

METHODS OF TEST

The cassava granulator was tested based on the AMTEC – Methods of Test.

DESCRIPTION OF THE MACHINE

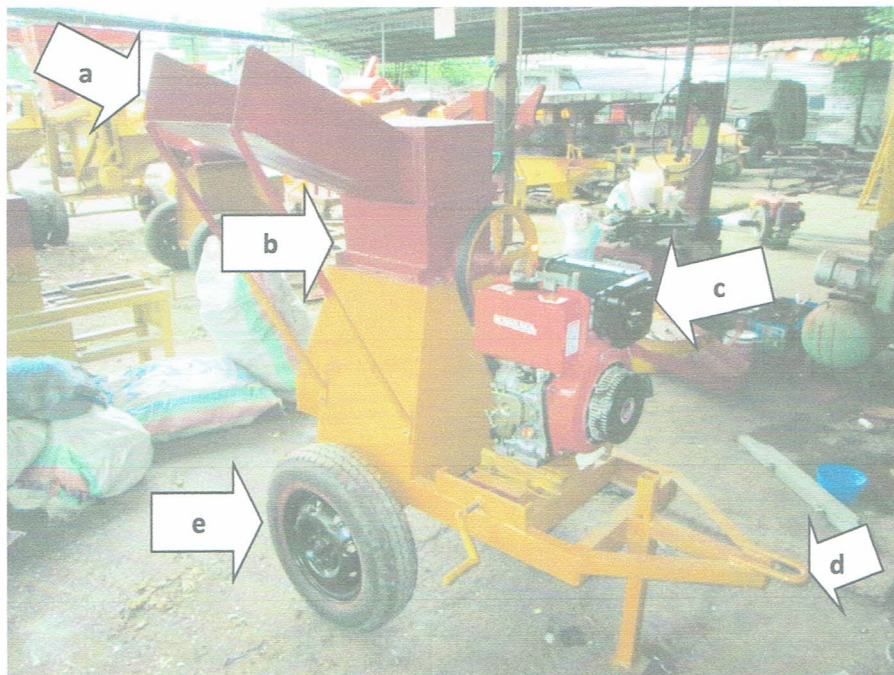


Figure 1. Parts of Cassava Granulator CJ G 1103 powered by 6.3 kW KAWAMA Diesel Engine:
(a) Loading Hopper, (b) Granulating chamber, (c) Prime mover, (d) Hitch point, and (e) Pneumatic tires

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification ^b
1 Main structure		
1.1 Overall dimensions, mm		
1.1.1 Length	ND	1565
1.1.2 Width	ND	1150
1.1.3 Height	ND	1570
1.2 Weight without prime mover, kg, if applicable	ND	NM
2 Loading hopper		
2.1 Material	ND	1.1 mm GI sheet
2.2 Bottom opening, mm		
2.2.1 Length	ND	150
2.2.2 Width	ND	245
2.3 Height from the ground, mm	ND	1430
3 Granulating mechanism		
3.1 Blade cover		
3.2 Type	ND	Cylindrical with louvers
3.3 Materials	ND	GI sheet
3.4 Blade		
3.4.1 Material	ND	Steel plate
3.4.2 Dimensions, mm		
3.4.2.1 Length	ND	130
3.4.2.2 Width	ND	65
3.4.2.3 Thickness	ND	11
3.4.2.4 Shape	ND	Bladed drum-type
3.4.2.5 Number of blades	ND	4
3.4.2.6 Clearance to the granulating cylinder, mm	ND	10
3.5 Grills		
3.5.1 Location	ND	Below the granulating cylinder
3.5.2 Material	ND	Round bar
3.5.3 Size, D, mm	ND	7

Note: ^aMachine specifications were not provided by the manufacturer.

^bBased on actual test and on-site measurement

ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification ^b
4	Output chute		
4.1	Material	ND	GI sheet
4.2	Dimensions, mm		
4.2.1	Length	ND	265
4.2.2	Diameter/width	ND	85
4.3	Height from the ground	ND	470
5	Safety devices	ND	ND
6	Special features	ND	Hitch point
7	Prime mover		
7.1	Brand	ND	KAWAMA
7.2	Model	ND	ND
7.3	Make or manufacturer	ND	Thailand
7.4	Serial number	ND	ND
7.5	Type	ND	Four-stroke, single vertical cylinder, diesel engine
7.6	Rated power, kW	ND	6.3
7.7	Rated speed, rpm	ND	3600
7.8	Cooling system	ND	Air-cooled
7.9	Starting system	ND	Rope-recoil
7.10	Weight, kg	ND	ND
8	Transmission system		
8.1	Type	ND	Belt & Pulley
9	Transport wheels		
9.1	Size	ND	145R12

Note: ^aMachine specifications were not provided by the manufacturer.

^bBased on actual test and on-site measurement

ND – No Data

RESULTS

Item		Data	
1	Condition of test materials		
1.1	Crop	Unpeeled cassava tubers	
1.2	Variety	ND	
1.3	Source	ND	
1.4	Weight, kg of input	250	
1.5	Moisture content, % _{wb}	NM	
1.6	Dimensions, mm		
1.6.1	Length	301	
1.6.2	Thickness	84.4	
2	Performance test		
2.1	Weight of output, kg	ND	
2.2	Operating time, h	0.097	
2.3	Input capacity, kg/h	2586.2	
2.4	Sieve analysis of 100g output sample, %		
2.4.1	Sieve #1/2	47.5	
2.4.2	Sieve #1/4	11.6	
2.4.3	Sieve #3/16	14.0	
2.4.4	Pan	26.9	
2.5	Speed of components, rpm	Without load	With load
2.5.1	Prime mover	3184	3175
2.5.2	Granulating shaft	685	677
2.6	Noise level, dB(A)	Without load	With load
2.6.1	Feeding operator	95.7	95.3
2.6.2	Bagger	94.2	96.4
2.7	Fuel consumption		
2.7.1	Fuel consumed, mL	130	
2.7.2	Fuel consumption rate, L/h	1.18	

Note: ND – No Data

Prepared by:


CONRAD IAN S. TAN
Test Engineer

Verified by:


ROMULO E. EUSEBIO
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

AUG 08 2019
Date

REMINDER

This test report should not be used for advertisement or for promotion of the product but only for evaluation and verification purposes and should not be reproduced in whole or in part without the permission of AMTEC.

For inquiries regarding the test report, you can visit us at our office, contact us at (+63) 921 400 7137 or (+63) 977 204 7236, send us a message at amtec.uplb@up.edu.ph or visit our official website at www.amtec.ceat.uplb.edu.ph.